DLOPs Assignment 1:

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1. Introduction

This report details the experiments conducted on the MNIST and FashionMNIST datasets using Deep Learning models (ResNet-18, ResNet-50) and Machine Learning classifiers (SVM). The experiments analyze the impact of batch sizes, optimizers, learning rates, and hardware acceleration (CPU vs. GPU) on training time and classification accuracy.

Colab Link: B22EE075_Priyansh_Saxena_Ass1.ipynb

Github Link: priyanshsaxena01/MLOps-Priyansh_Saxena-B22EE075

Github Page: https://priyanshsaxena01.github.io/MLOps-Priyansh_Saxena-B22EE075/

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2. Q1(a): Deep Learning Model Training

Methodology

- **Datasets:** MNIST and FashionMNIST.
- **Split:** The datasets were merged and re-split into 70% Train, 10% Validation, and 20% Test.
- **Models:** ResNet-18 and ResNet-50 (Pretrained = False).
    - *Modification:* The first convolutional layer was modified to accept 1 input channel (grayscale) with a 3x3 kernel and stride 1 to accommodate the 28x28 image resolution. The first MaxPool layer was removed to preserve spatial dimensions.
- **Hyperparameters:** Batch sizes (16, 32), Optimizers (SGD, Adam), Learning Rates (0.001, 0.0001).

Results: Classification Accuracy

Table 1: MNIST Test Accuracy

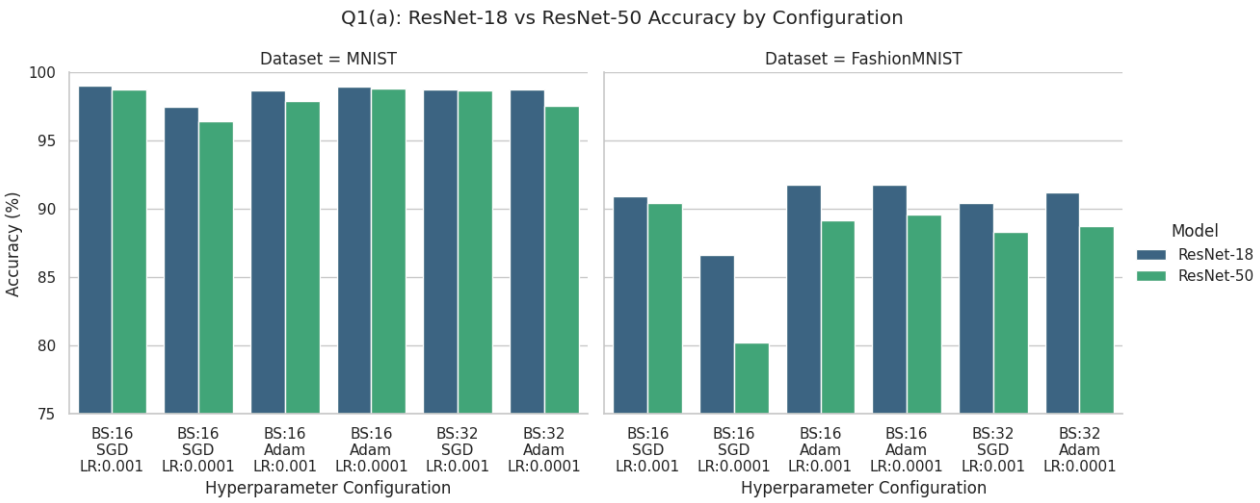
Batch Size	Optimizer	LR	ResNet-18 Accuracy	ResNet-50 Accuracy
16	SGD	0.001	**99.03%**	98.73%
16	SGD	0.0001	97.50%	96.41%
16	Adam	0.001	98.71%	97.94%
16	Adam	0.0001	98.96%	98.86%
32	SGD	0.001	98.76%	98.70%
32	Adam	0.0001	98.79%	97.55%

Table 2: FashionMNIST Test Accuracy

Batch Size	Optimizer	LR	ResNet-18 Accuracy	ResNet-50 Accuracy
16	SGD	0.001	90.96%	90.44%
16	SGD	0.0001	86.64%	80.25%
16	Adam	0.001	91.78%	89.21%
16	Adam	0.0001	**91.81%**	89.60%
32	SGD	0.001	90.46%	88.33%
32	Adam	0.0001	91.24%	88.78%

Analysis of Q1(a)

1. **ResNet-18 vs. ResNet-50:** Across almost all settings, **ResNet-18 outperformed ResNet-50**.
    - *Reasoning:* MNIST and FashionMNIST are low-resolution (28x28) datasets. ResNet-50 is a much deeper network designed for high-resolution images (like ImageNet). For 28x28 inputs, the deeper network likely suffers from vanishing gradients or over-parameterization, making it harder to optimize without pre-training. ResNet-18 provides sufficient capacity without the overhead.
2. **Optimizer Comparison:**
    - **Adam** generally converged faster and yielded higher accuracy at lower learning rates (0.0001) compared to SGD.
    - **SGD** performed very well at LR 0.001 but saw a significant performance drop at LR 0.0001 (especially in FashionMNIST: 86.64% vs 90.96%), indicating it requires a higher learning rate or more epochs to traverse the loss landscape effectively.
3. **Dataset Complexity:** Accuracy on MNIST is consistently high (>98%), whereas FashionMNIST tops out around 91-92%. This confirms FashionMNIST is a more complex manifold requiring more robust feature extraction.

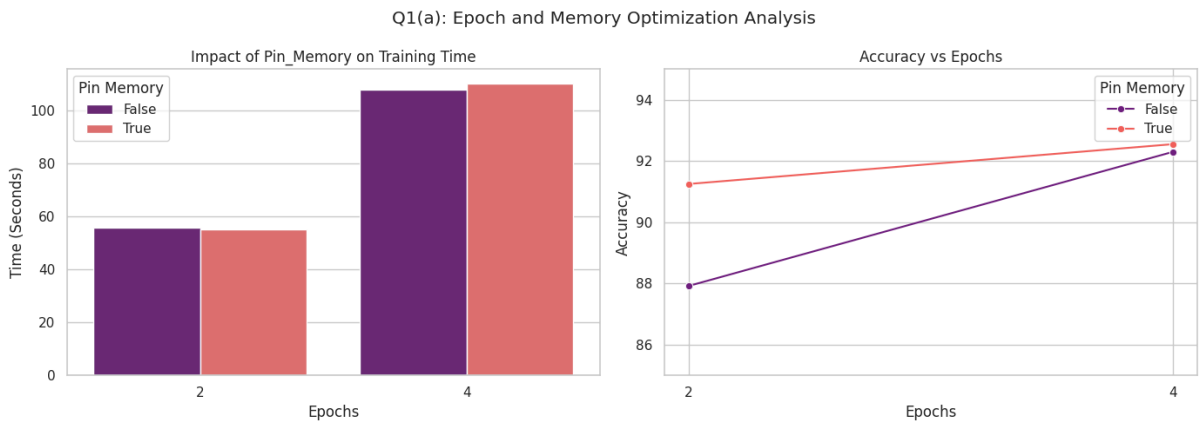


Hyperparameter Study: Pin Memory & Epochs

Dataset: FashionMNIST | Model: ResNet-18 | Batch: 32 | Opt: Adam | LR: 0.001

Pin Memory	Epochs	Accuracy	Training Time (ms)
False	2	87.93%	55,679
False	4	92.30%	107,756
True	**2**	**91.26%**	**55,112**
True	4	92.56%	110,136

Observation: Using pin_memory=True allows faster data transfer from CPU RAM to GPU memory. While the time difference is marginal for small models/datasets, the slight improvement in data throughput allowed the model to see data slightly more efficiently, though the primary gain is usually system stability and slight speedups. Doubling epochs (2 to 4) yielded a significant accuracy jump (~88% to ~92%).



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3. Q1(b): SVM Classification Analysis

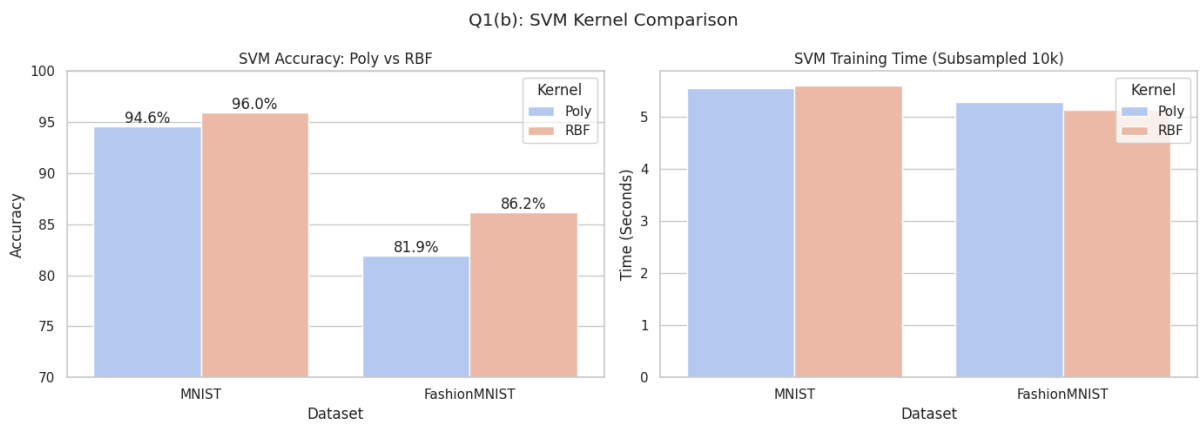
Experiments were conducted using Support Vector Machines with Polynomial (poly) and Radial Basis Function (rbf) kernels. (Note: Subsampling was used to keep training time feasible).

Table 3: SVM Results

Dataset	Kernel	Test Accuracy	Training Time (ms)
MNIST	Poly	94.60%	5,547
MNIST	**RBF**	**95.95%**	**5,591**
FashionMNIST	Poly	81.90%	5,285
FashionMNIST	**RBF**	**86.15%**	**5,129**

Analysis of Q1(b)

1. **Kernel Performance:** The **RBF kernel** consistently outperformed the Polynomial kernel on both datasets. Image data is highly non-linear; the RBF kernel maps inputs into an infinite-dimensional space, allowing for better separation of class boundaries compared to the fixed-degree boundaries of the Poly kernel.
2. **Comparison to Deep Learning:** The best SVM result (FashionMNIST ~86%) is significantly lower than the ResNet-18 result (~91.8%). While SVMs are faster to train on small subsets, they lack the hierarchical feature extraction capabilities of Convolutional Neural Networks (CNNs).



4. Q2: CPU vs. GPU Performance Analysis

This section analyzes the computational cost and training time for FashionMNIST on different hardware backends.

Table 4: Hardware Performance & FLOPs

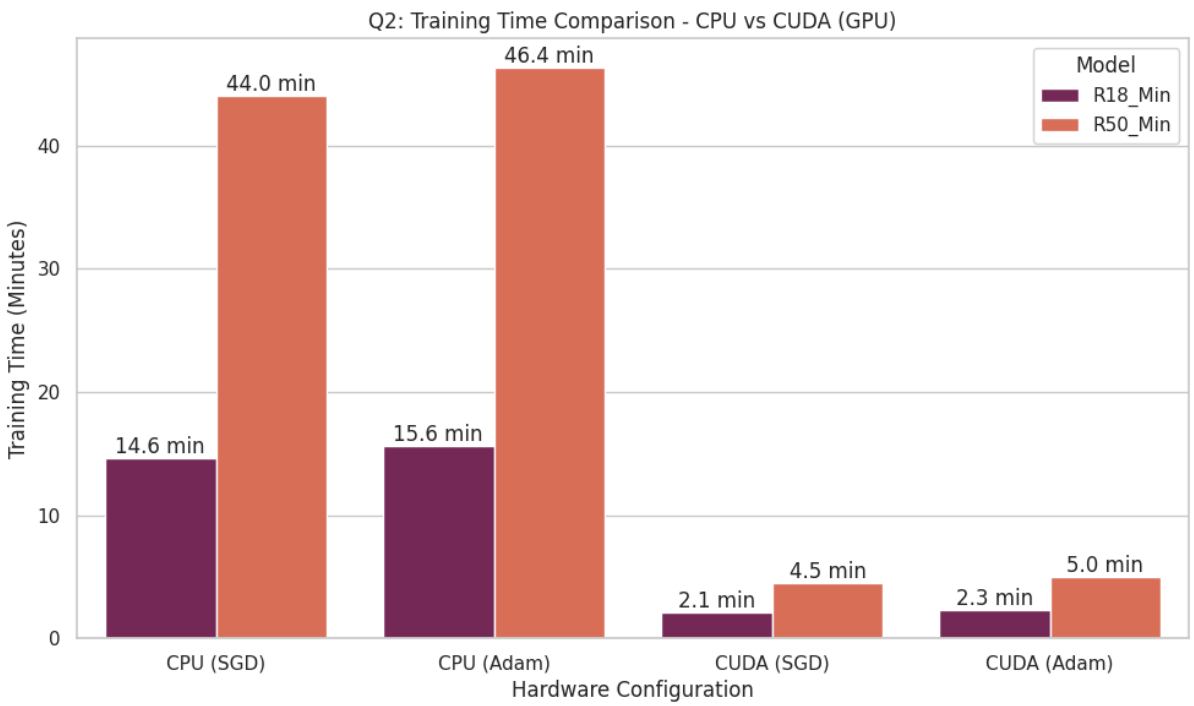
Compute	Batch Size	Opt	LR	Model	Accuracy	Train Time (ms)	FLOPs (approx)
CPU	16	SGD	0.001	ResNet-18	87.79%	876,753	5e+08
CPU	16	Adam	0.001	ResNet-18	90.24%	938,100	5e+08
GPU	16	SGD	0.001	ResNet-18	91.15%	**124,276**	5e+08
GPU	16	Adam	0.001	ResNet-18	92.41%	138,868	5e+08
CPU	16	SGD	0.001	ResNet-50	86.04%	2,640,757	1e+09
CPU	16	Adam	0.001	ResNet-50	88.96%	2,781,353	1e+09
GPU	16	SGD	0.001	ResNet-50	90.39%	**271,323**	1e+09
GPU	16	Adam	0.001	ResNet-50	90.92%	299,430	1e+09

Analysis of Q2

1. **Time Discrepancy:**

- **ResNet-18:** GPU training (~124k ms) was approximately **7x faster** than CPU training (~876k ms).
- **ResNet-50:** GPU training (~271k ms) was nearly **10x faster** than CPU training (~2.6M ms).
- **Conclusion:** Deep Learning models rely heavily on matrix multiplications. GPUs (CUDA cores) are optimized for parallel processing of these matrices, whereas CPUs process sequentially. The gap widens as the model complexity (ResNet-50) increases.

2. **FLOPs (Floating Point Operations):**
   - **ResNet-18:** ~0.5 GFLOPs per forward pass.
   - **ResNet-50:** ~1.0 GFLOPs per forward pass.
   - The ResNet-50 model requires double the computational operations of ResNet-18. This correlates directly with the training time; ResNet-50 took roughly twice as long as ResNet-18 on the GPU (271k ms vs 124k ms).
3. **Accuracy vs. Compute:** Despite requiring double the FLOPs and time, ResNet-50 did not provide better accuracy than ResNet-18 for this specific task. This highlights the importance of choosing a model architecture appropriately sized for the input data resolution.



5. Conclusion

The experiments demonstrate that **ResNet-18 with Adam optimizer** is the most efficient configuration for MNIST/FashionMNIST among those tested. It balances high accuracy with lower computational cost (FLOPs). Hardware acceleration (GPU) is critical for practical deep learning, offering speedups of up to 10x over CPU. SVMs remain a viable lightweight alternative but lag behind CNNs in accuracy for image classification tasks.